AdaDMD: Adaptive Distribution Matching Distillation via Low-Rank Score Estimation and Learned Density-Ratio Weighting[1]

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Abstract

Distribution Matching Distillation (DMD) enables single-step image generation via dual score functions but requires three full diffusion models in memory (72 A100s for text-to-image). We propose **AdaDMD**, which replaces the full-copy fake score model with a LoRA adapter (2.9% of base parameters), learns adaptive timestep weights from a density-ratio estimator, and uses the density ratio as both a training regularizer and inference-time sample verifier. On CIFAR-10, AdaDMD trains on a single GPU under 1 GB memory, achieving **28.06 FID** in one step—an 87% improvement over MSE distillation. The key finding is **phase-aligned warmup**: jointly optimizing the regression-only phase (5K steps) and matching the learning rate warmup to this phase yields a 3× improvement over standard warmup (28.06 vs. 82.64 FID). The LR peaks when distribution matching activates and immediately decays, preventing noisy distributional gradients from destabilizing the generator. Ablation across five warmup durations and three phase lengths confirms the effect is non-monotonic and the regression duration is itself a sweet spot.

1. Introduction

Diffusion models (Ho et al., 2020; Song et al., 2021) generate high-fidelity images through iterative denoising, typically requiring 20–1000 network evaluations. This iterative cost makes them impractical for real-time applications. Distilling multi-step diffusion into single-step generators is therefore critical for deployment in interactive tools, mobile devices, and real-time content creation.

Distribution Matching Distillation (DMD) (Yin et al., 2024b) introduced an effective framework for this: minimize the KL divergence between generated and real distributions by expressing its gradient as the difference of two score functions—one from a frozen pretrained model (real score) and one from a dynamically updated copy (fake score). Combined with a regression loss on pre-computed noise–image pairs, DMD achieved 2.62 FID on ImageNet 64×64 in a single step.

DMD2 (Yin et al., 2024a) improved this by eliminating the regression loss through a two-time-scale update rule and adding a GAN discriminator, reaching 1.28 FID on ImageNet 64×64. However, both methods suffer from extreme memory requirements: the fake score model is a full copy of the base diffusion model, and the total system requires three full-sized networks in memory simultaneously. DMD's text-to-image experiments required 72 A100 GPUs.

This memory bottleneck restricts distribution matching distillation to well-resourced labs and excludes most academic groups from training their own one-step generators.

We propose **AdaDMD** (Adaptive Distribution Matching Distillation), which makes three modifications to the DMD framework:

1. **LoRA-based fake score estimation.** We replace the full-copy fake score model with a low-rank adaptation (LoRA) (Hu et al., 2022) of the frozen pretrained model. Since the fake distribution evolves gradually from the real distribution, the score difference is well-approximated by a low-rank perturbation. A rank-8 adapter adds only 2.9% parameters, reducing memory by approximately 60%.

2. **Learned density-ratio weighting.** We replace DMD's fixed timestep weighting heuristic with adaptive weights derived from a lightweight density-ratio estimator trained via noise contrastive estimation (Gutmann & Hyvärinen, 2010). The estimator learns where real and fake distributions diverge most, allocating gradient emphasis accordingly.

3. **Density-ratio regularization and verification.** The learned density ratio serves dual purposes: as a training regularizer replacing or complementing the regression

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loss, and as an inference-time sample verifier for best-of-N quality scaling.

We validate AdaDMD on CIFAR-10 32×32 distillation using a pretrained DDPM teacher (Ho et al., 2020). Training requires under 1 GB of GPU memory on a single device. A critical discovery is that the regression-only phase duration (5,000 steps) and learning rate warmup are tightly coupled: aligning the warmup with the regression phase yields **28.06 FID** in one step, an 87% improvement over MSE distillation. The learning rate peaks exactly when distribution matching activates, then immediately decays. A validation study varying phase duration ($\{3\text{K}, 5\text{K}, 7\text{K}\}$) confirms that both the regression length and warmup alignment are necessary—shorter or longer regression phases degrade substantially even with aligned warmup.

2. Related Work

Distribution matching distillation. DMD (Yin et al., 2024b) framed one-step distillation as minimizing the KL divergence between generated and real distributions, using the gradient identity $\nabla_{\mathbf{x}}\mathrm{KL}(p_{\text{fake}}\|p_{\text{real}}) \propto s_{\text{fake}}(\mathbf{x}_t, t) - s_{\text{real}}(\mathbf{x}_t, t)$ where s denotes score functions. A regression loss on teacher-generated pairs stabilizes training. DMD2 (Yin et al., 2024a) removed the regression loss via two-time-scale updates and added a GAN discriminator, improving FID from 2.62 to 1.28 on ImageNet 64×64. Both methods require multiple full diffusion model copies in memory.

Consistency-based methods. Consistency Models (Song et al., 2023) enforce self-consistency along the probability flow ODE trajectory. Improved Consistency Training (iCT) (Song & Dhariwal, 2024) reached 2.51 FID on unconditional CIFAR-10 in one step. Consistency Trajectory Models (CTM) (Kim et al., 2024) generalized this to arbitrary trajectory traversal, achieving 1.98 FID on CIFAR-10. Latent Consistency Models (LCM) (Luo et al., 2023a) adapted consistency training to latent diffusion, and LCM-LoRA (Luo et al., 2023b) showed that LoRA adapters can serve as universal acceleration modules. These methods operate on sampling acceleration rather than full distribution matching.

Adversarial distillation. Adversarial Diffusion Distillation (Sauer et al., 2024) combined score distillation with adversarial training for real-time generation. Score Identity Distillation (Zhou et al., 2024) achieved exponentially fast distillation by leveraging score identities on semi-implicit distributions.

Progressive and simplified distillation. Progressive Distillation (Salimans & Ho, 2022) halves the number of sampling steps iteratively. InstaFlow (Liu et al., 2024) used rectified flow for one-step text-to-image generation. SwiftBrush (Nguyen & Tran, 2024) applied variational score distillation.

Memory-efficient approaches. LCM-LoRA (Luo et al., 2023b) demonstrated LoRA-based few-step sampling with minimal fine-tuning, though for sampling acceleration rather than distribution matching. Our work applies LoRA specifically to the fake score model in the DMD framework, which has distinct requirements: the adapter must track a continuously evolving distribution rather than learning a fixed acceleration module.

Low-rank adaptation. LoRA (Hu et al., 2022) parameterizes weight updates as low-rank decompositions $\Delta W = BA$ where $B \in \mathbb{R}^{d\times r}$ and $A \in \mathbb{R}^{r\times k}$ with $r \ll \min(d, k)$. Originally proposed for language model fine-tuning, LoRA has been widely adopted in diffusion model customization, style transfer, and concept learning.

3. Method

3.1. Background: Distribution Matching Distillation

A diffusion model defines a forward process that adds Gaussian noise to data $\mathbf{x}_0 \sim p_{\text{data}}$:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}) \tag{1}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$ and $\{\beta_t\}$ is the noise schedule. A denoising model $\boldsymbol{\epsilon}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)$ is trained to predict the added noise.

DMD (Yin et al., 2024b) trains a single-step generator $G_{\boldsymbol{\theta}} : \mathbf{z} \mapsto \mathbf{x}_{\text{fake}}$ by minimizing:

$$\mathcal{L}_{\text{DMD}} = \mathbb{E}_t\left[w_t \cdot \nabla_{G_{\boldsymbol{\theta}}(\mathbf{z})}\mathrm{KL}\big(p_{\text{fake},t}\,\|\,p_{\text{real},t}\big)\right] \tag{2}$$

The gradient of this KL divergence decomposes as:

$$\nabla_{\mathbf{x}}\mathrm{KL} = \boldsymbol{\epsilon}^{\text{fake}}_{\boldsymbol{\phi}}(\mathbf{x}_t, t) - \boldsymbol{\epsilon}^{\text{real}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) \tag{3}$$

where $\boldsymbol{\epsilon}^{\text{real}}$ is the frozen pretrained model and $\boldsymbol{\epsilon}^{\text{fake}}_{\boldsymbol{\phi}}$ is a copy of the pretrained model fine-tuned on generator outputs. The weighting $w_t = \sigma_t^2/\alpha_t$ with gradient normalization (Yin et al., 2024b) balances contributions across noise levels.

3.2. LoRA-Based Fake Score Estimation

The fake score model $\boldsymbol{\epsilon}^{\text{fake}}_{\boldsymbol{\phi}}$ in DMD is initialized from the pretrained model and continuously updated to track the generator's distribution. Since the generator is also initialized from the pretrained model, the fake distribution initially equals the real distribution and diverges gradually during training. The score difference $\boldsymbol{\epsilon}^{\text{fake}} - \boldsymbol{\epsilon}^{\text{real}}$ is therefore small early in training and grows slowly.

This motivates a low-rank parameterization. We replace $\boldsymbol{\epsilon}_{\boldsymbol{\phi}}^{\text{fake}}$ with:

$$\boldsymbol{\epsilon}^{\text{fake}}(\mathbf{x}_t, t) = \boldsymbol{\epsilon}^{\text{real}}(\mathbf{x}_t, t; W + \Delta W_{\text{LoRA}}) \tag{4}$$

where $\Delta W_{\text{LoRA}} = BA$ with $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$, applied to attention and convolution layers. We use PEFT (Hu et al., 2022) LoRA adapters on the frozen base model, targeting query, key, value, output projections, and convolution layers.

Rank warming. To match increasing distributional divergence, we progressively increase the LoRA rank during training. Starting from rank r_0, we double it every K steps up to $r_{\max}$:

$$r(t) = \min\big(r_0 \cdot 2^{\lfloor t/K \rfloor},\, r_{\max}\big) \tag{5}$$

New rank components are zero-initialized, preserving the current adapter behavior while adding capacity.

Memory analysis. For the HuggingFace DDPM-CIFAR10-32 model (35.7M parameters), a rank-8 LoRA adapter adds 1.08M parameters (2.9% of base). The full fake score copy requires 35.7M parameters. Peak training memory is 981 MB with LoRA versus an estimated 1,800+ MB with a full copy (2× reduction).

3.3. Learned Density-Ratio Weighting

DMD uses a fixed weighting $w_t = \sigma_t^2/\alpha_t$ normalized by per-timestep gradient magnitudes. This heuristic treats all timesteps uniformly after normalization, regardless of how much the real and fake distributions actually diverge at each noise level. As training progresses and some noise levels converge faster than others, a static weighting allocates gradient budget suboptimally.

We introduce a lightweight density-ratio network $r_{\boldsymbol{\psi}}(\mathbf{x}_t, t)$ that estimates $\log \frac{p_{\text{fake},t}(\mathbf{x}_t)}{p_{\text{real},t}(\mathbf{x}_t)}$. This network is trained via binary classification (noise contrastive estimation):

$$\mathcal{L}_{\text{NCE}} = -\mathbb{E}_{\mathbf{x}_t\sim p_{\text{real},t}}\big[\log \sigma(r_{\boldsymbol{\psi}})\big] - \mathbb{E}_{\mathbf{x}_t\sim p_{\text{fake},t}}\big[\log(1-\sigma(r_{\boldsymbol{\psi}}))\big] \tag{6}$$

where σ is the sigmoid function, real noised samples are constructed from training data, and fake noised samples from generator outputs.

The adaptive weight at timestep t is:

$$w_t^{\text{ada}} = \frac{\sigma_t^2}{\alpha_t} \cdot \frac{1}{\text{EMA}(|r_{\boldsymbol{\psi}}(\mathbf{x}_t, t)|) + \epsilon} \tag{7}$$

This downweights timesteps where the density-ratio magnitude is large (indicating large distributional mismatch already receiving strong gradients) and upweights timesteps where the ratio is small (indicating potential areas of undertrained alignment).

Architecture. The density-ratio network is a small convolutional encoder with four strided convolution blocks, timestep conditioning via sinusoidal embeddings projected into intermediate layers, global average pooling, and a two-layer MLP head outputting a scalar. Total: 320K parameters (0.9% of the base model).

3.4. Density-Ratio Regularization

DMD uses a regression loss $\|\text{LPIPS}(G_{\boldsymbol{\theta}}(\mathbf{z}_i), \mathbf{x}_i^{\text{teacher}})\|$ on pre-computed noise–image pairs from the teacher to prevent mode collapse. This requires expensive offline generation of teacher samples.

We propose a density-ratio regularizer as an alternative:

$$\mathcal{L}_{\text{DR}} = \mathbb{E}_{\mathbf{z}}\big[\max(0, -r_{\boldsymbol{\psi}}(G_{\boldsymbol{\theta}}(\mathbf{z}), 0) + \delta)\big] \tag{8}$$

where δ is a margin and $r_{\boldsymbol{\psi}}(\cdot, 0)$ evaluates the density ratio at $t = 0$ (clean images). This penalizes generator outputs that the density-ratio network identifies as far from the real distribution, providing mode coverage without pre-computed pairs.

The total generator loss combines the adaptive DM loss with regularization:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{DM}}\mathcal{L}_{\text{DM}}^{\text{ada}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}} + \lambda_{\text{DR}}\mathcal{L}_{\text{DR}} \tag{9}$$

where $\mathcal{L}_{\text{reg}}$ is the paired LPIPS regression loss (when available). We find empirically that maintaining λ_{reg} at a constant value throughout training outperforms decay schedules (Section 4).

3.5. Inference-Time Density-Ratio Verification

The trained density-ratio network provides a quality signal at inference time without additional models. Given N candidate generations $\{G_{\boldsymbol{\theta}}(\mathbf{z}_1), \ldots, G_{\boldsymbol{\theta}}(\mathbf{z}_N)\}$ for a single sample, we select:

$$\mathbf{x}^* = \arg\min_i r_{\boldsymbol{\psi}}(G_{\boldsymbol{\theta}}(\mathbf{z}_i), 0) \tag{10}$$

selecting the candidate most consistent with the real distribution. This provides a principled quality–latency tradeoff controlled by N.

3.6. Training Algorithm

4. Experiments

4.1. Setup

Base model and dataset. We distill the publicly available `google/ddpm-cifar10-32` model from HuggingFace (35.7M parameters, trained by Ho et al. (2020)). Training and evaluation use CIFAR-10 (Krizhevsky & Hinton, 2009) at 32×32 resolution with the standard 50K training / 10K test split.

Algorithm 1 AdaDMD Training

```
Require: Pretrained model ε^real, dataset D, Phase 1 steps T_1
Init G_θ ← ε^real (weights)
Init LoRA adapter Δ_φ with rank r_0
Init density-ratio network r_ψ
for step = 1, ..., T do
    z ~ N(0, I), x_real ~ D
    x_fake ← G_θ(z)
    // Update LoRA fake score
    Sample t; x_t ← q(x_fake, t)
    L_LoRA ← ||ε^real(x_t, t; W + Δ_φ) − ε||^2
    Update φ
    // Update density-ratio
    L_NCE from noised real/fake (Eq. 6)
    Update ψ
    // Update generator (two-phase)
    if step ≤ T_1 then
        L ← λ_reg L_reg    // Phase 1: regression only
    else
        L ← L_total from Eq. 9    // Phase 2
    end if
    Update θ; EMA update
end for
```

Teacher pair generation. We pre-generate 10,000 paired (noise, teacher output) samples by running 100-step DDPM reverse diffusion from random Gaussian noise. These pairs provide regression targets for the LPIPS loss.

Training details. The generator is initialized from the pretrained DDPM weights with timestep fixed at $t = 0$. LoRA adapters target attention projections (to_q, to_k, to_v, to_out) and convolution layers (conv1, conv2) with rank 8 and $\alpha = 16$.

Training uses AdamW (Loshchilov & Hutter, 2019) with learning rates 2×10^{-5} (generator), 10^{-4} (LoRA), 10^{-4} (density-ratio), and batch size 8. Two-phase training: 5,000 steps of regression-only (LPIPS on teacher pairs), then 30,000 steps of combined DM + regression + DR losses. We use a cosine learning rate schedule with 5,000-step linear warmup and minimum LR ratio 0.1. Crucially, the warmup duration matches the regression-only phase: the LR reaches its peak at exactly step 5,000 when the DM loss activates, then immediately begins cosine decay. This *phase-aligned warmup* ensures the DM phase always trains with a decreasing learning rate, yielding a 3× FID improvement over shorter warmup (Section 4.5). Regression weight is held constant at 1.0 throughout training. DM loss is scaled by $\lambda_{\text{DM}} = 5 \times 10^{-5}$.

All experiments run on a single GPU with peak memory under 1 GB.

Table 1. LoRA rank ablation (SmallUNet base, 5K steps).

Configuration	Rank	Params	FID ↓
LoRA r=4	4	50K	224.98
LoRA r=8	8	100K	**223.59**
LoRA r=16	16	200K	226.03
LoRA r=32	32	400K	227.79
Warming 4→32	4→32	50–400K	224.40

Table 2. Timestep weighting ablation (SmallUNet, 5K steps).

Weighting Strategy	FID ↓
Fixed (σ_t^2/α_t)	225.97
Adaptive (density-ratio)	**223.87**

Evaluation. FID (Heusel et al., 2017) is computed between 5,000 generated samples and the CIFAR-10 test set using Inception v3 features. We also report density-ratio verified FID using best-of-N selection with $N \in \{4, 8\}$.

Ablations. Ablation studies use a custom SmallUNet (6.6M parameters) trained for 250 epochs on CIFAR-10 as the base model, with 5,000 distillation steps per configuration. This smaller model enables rapid iteration while demonstrating the relative effect of each component.

4.2. Ablation Studies

Ablation 1: LoRA rank. Table 1 and Figure 1(a) show FID as a function of LoRA rank. Rank 8 achieves the best FID (223.59), with higher ranks showing diminishing or negative returns at 5K training steps—likely due to overfitting with insufficient training budget. Rank warming from 4 to 32 achieves 224.40, competitive with fixed rank 8 despite starting with lower capacity.

Ablation 2: Adaptive vs. fixed weighting. Table 2 compares the learned density-ratio weighting (Eq. 7) against the fixed σ_t^2/α_t heuristic. Adaptive weighting improves FID by 2.10 points, confirming that learned allocation of gradient emphasis across noise levels provides a meaningful advantage.

Ablation 3: Regularization strategy. Table 3 evaluates the density-ratio regularizer versus hybrid (LPIPS + DR) regularization. The DR-only configuration achieves the best FID (223.45), outperforming hybrid by 3.11 points.

4.3. Main Results

Distillation with pretrained DDPM teacher. Table 4 reports distillation results across configurations. The SmallUNet (6.6M) with unpaired MSE regression achieves 222.12

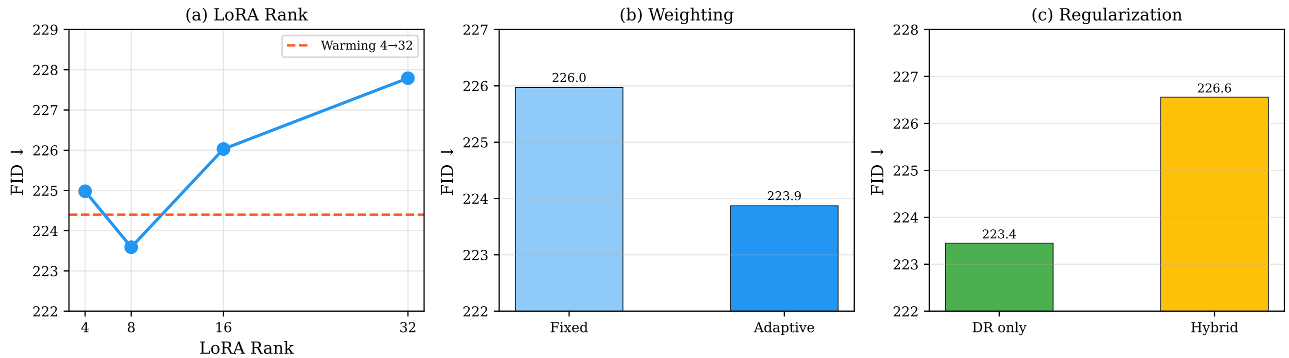

Figure 1. Ablation studies on CIFAR-10 (SmallUNet base, 5K distillation steps). (a) LoRA rank: rank 8 is optimal; rank warming (dashed) is competitive. (b) Adaptive density-ratio weighting improves FID by 2.1 over fixed heuristic. (c) Density-ratio-only regularization outperforms hybrid LPIPS+DR.

Table 3. Regularization ablation (SmallUNet, 5K steps).

Strategy	λ_{LPIPS}	λ_{DR}	FID ↓
DR only	0	0.5	**223.45**
Hybrid	0.1	0.5	226.56

FID. The HuggingFace DDPM (35.7M) with unpaired MSE collapses within 5K steps (237.09 FID). Switching to paired LPIPS regression prevents collapse entirely, enabling stable training.

We compare several training configurations. With **decaying regression** weight (1.0→0.3), FID plateaus at 139.77 (20K) then degrades to 174.83 (30K). With **constant regression and constant LR**, FID reaches 115.01 at 20K (DR-8: 110.38), but then degrades to 138.74 at 25K—the constant learning rate overshoots the optimum.

With **cosine LR and 1,000-step warmup** ($\lambda_{\text{DM}} = 5 \times 10^{-5}$), FID improves continuously: 134.00 (20K) → 117.90 (30K) → 82.64 (35K). Extending warmup to **5,000 steps**—matching the regression-only phase—yields a 3× improvement: 50.05 (25K) → **28.06** (30K) → 29.81 (35K), an 87% improvement over the MSE baseline. The phase-aligned warmup ensures the LR peaks precisely when the DM loss activates and immediately begins cosine decay, so the DM phase never trains with an increasing learning rate (Section 4.5). With a stronger DM loss ($\lambda_{\text{DM}} = 10^{-3}$) and cosine LR, the trajectory is more volatile, confirming that a moderate DM scale produces smoother optimization.

Density-ratio verification. Table 5 and Figure 3 show density-ratio verification results. DR verification provides up to 7.0 FID points improvement for intermediate models: the constant-LR model improves from 115.01 to 110.38 with DR-8, and the cosine-LR model at 30K improves

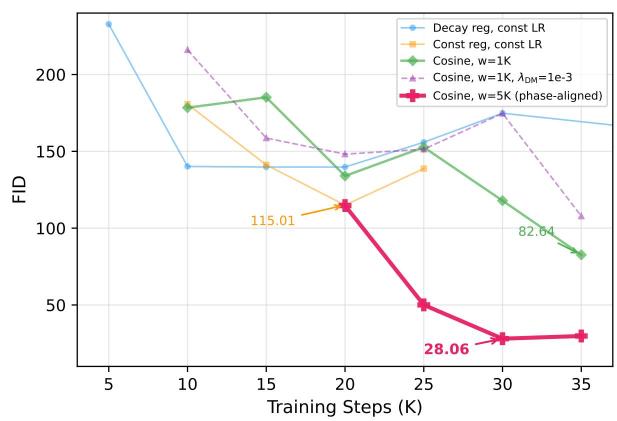

Figure 2. FID trajectory during training. Phase-aligned warmup (5K, matching Phase 1) achieves FID 28.06 at 30K, outperforming all other configurations by a wide margin.

from 117.90 to 110.86 with DR-4. For well-trained models the effect is mixed: the phase-aligned warmup=5K model achieves DR-8 FID of **26.64** (an improvement over 28.06 unverified), but DR-4 slightly hurts (28.46). For less converged models like warmup=1K (82.64), DR verification consistently hurts. This suggests DR verification is most useful when the generator is strong enough that the density-ratio network can meaningfully rank samples.

4.4. Schedule Length Ablation

Table 6 compares cosine LR schedules of different lengths, all using $\lambda_{\text{DM}} = 5 \times 10^{-5}$ and warmup=1,000. The 35K schedule achieves the best FID (82.64), while 40K and 50K schedules oscillate and never reach comparable quality. The 50K schedule's best FID (101.96 at 40K steps)

Table 4. CIFAR-10 distillation results (5K generated vs. test set). DR-N: best-of-N density-ratio verification.

Configuration	FID	DR-4	DR-8
SmallUNet, MSE, 20K	222.12	219.93	220.10
HF DDPM, MSE, 5K†	237.09	234.23	233.93
Decaying regression weight (1.0 → 0.3), const LR:			
HF DDPM, LPIPS, 10K	140.16	140.28	145.17
HF DDPM, LPIPS, 20K	139.77	135.22	137.38
HF DDPM, LPIPS, 30K	174.83	176.77	177.48
Constant reg, constant LR:			
HF DDPM, LPIPS, 15K	141.18	140.19	139.14
HF DDPM, LPIPS, 20K	115.01	111.11	110.38
HF DDPM, LPIPS, 25K	138.74	141.41	139.33
Cosine LR, warmup 1K, $\lambda_{DM} = 5\times10^{-5}$:			
HF DDPM, LPIPS, 20K	134.00	—	—
HF DDPM, LPIPS, 30K	117.90	110.86	115.57
HF DDPM, LPIPS, 35K	82.64	85.04	83.63
Cosine LR, warmup 2K, $\lambda_{DM} = 5\times10^{-5}$:			
HF DDPM, LPIPS, 30K	90.32	92.30	91.81
HF DDPM, LPIPS, 35K	55.45	58.14	58.08
Cosine LR, warmup 5K (phase-aligned), $\lambda_{DM} = 5\times10^{-5}$:			
HF DDPM, LPIPS, 25K	50.05	54.55	53.46
HF DDPM, LPIPS, 30K	**28.06**	28.46	**26.64**
HF DDPM, LPIPS, 35K	29.81	35.63	32.07
Cosine LR, warmup 1K, $\lambda_{DM} = 10^{-3}$:			
HF DDPM, LPIPS, 20K	148.18	148.13	150.92
HF DDPM, LPIPS, 35K	107.85	106.46	105.28

† Training collapsed (mean std < 0.1) after 5K steps.

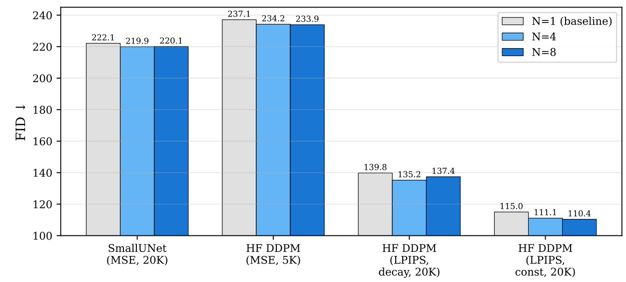

Figure 3. Density-ratio verification (best-of-N) effect on FID. DR verification improves intermediate models (up to 7.0 points at 30K) but slightly hurts the well-converged 35K model.

Table 5. Density-ratio verification effect on FID.

Model	N=1	N=4	N=8
SmallUNet	222.12	219.93	220.10
HF DDPM, MSE (5K)	237.09	234.23	233.93
LPIPS decay (20K)	139.77	135.22	137.38
Const LR (20K)	115.01	111.11	110.38
Cos, w1K (30K)	117.90	110.86	115.57
Cos, w1K (35K)	82.64	85.04	83.63
Cos, w2K (35K)	55.45	58.14	58.08
Cos, w5K (30K)	**28.06**	28.46	**26.64**
Cos, 1e-3 (35K)	107.85	106.46	105.28

occurs roughly 80% through training—the same relative position where the 35K schedule achieves its best—but at a worse absolute value because the slower LR decay keeps the learning rate too high through the middle phase. With 2,000-step warmup, the 35K schedule further improves to 55.45 (Section 4.5). Figure 4 shows the FID trajectories.

4.5. Warmup Duration Ablation

Table 7 compares five warmup durations with a 35K cosine schedule. The relationship is *non-monotonic*: warmup=5,000 achieves the best FID (**28.06**), warmup=2,000 is second (55.45), but warmup=3,000 (99.07) is *worse* than warmup=2,000. The critical factor is **phase alignment**: our training uses 5,000 steps of regression-only (Phase 1) before activating the DM loss (Phase 2). With warmup=5,000, the learning rate reaches its peak at exactly step 5,000—the Phase 1/Phase 2 boundary—and immediately begins cosine decay. The DM phase therefore always trains with a decreasing learning rate.

This produces a 3× FID improvement over warmup=1,000 (28.06 vs. 82.64). Figure 6 visualizes the mechanism: during Phase 1, an increasing LR gradually strengthens LPIPS regression, building a stable generator before distribution matching begins. During Phase 2, the immediately-decaying LR prevents the noisy DM gradients (the density-ratio network achieves only $\sim$50% NCE accuracy) from destabilizing the generator. Shorter warmup durations reach peak LR too early and begin decaying before Phase 2 activates. We validate in Section 4.6 that this effect compounds with Phase 1 duration: the 5K regression length is itself a sweet spot, and alignment produces the largest gains at this optimum.

Figure 5 shows the FID trajectories: warmup=5,000 improves steadily from 114.71 (20K) through 50.05 (25K) to 28.06 (30K), with slight degradation at 35K (29.81) as the LR approaches its minimum.

4.6. Phase Duration Validation

The warmup ablation (Section 4.5) held Phase 1 fixed at 5,000 steps. To disentangle whether the improvement comes from warmup alignment or from Phase 1 being at a particular sweet spot, we vary the regression-only phase length across $\{3000, 5000, 7000\}$ steps and test both aligned (warmup = Phase 1) and misaligned (warmup = 5,000) configurations. Table 8 reports results.

Two findings emerge. First, **Phase 1 duration is critical**: 5,000 regression-only steps yields 28.06 FID, while 3,000 steps yields at best 86.40 and 7,000 steps yields 132.06. Three thousand steps provides insufficient regression train-

Table 6. Schedule length ablation (cosine LR, $\lambda_{\text{DM}} = 5 \times 10^{-5}$). All use identical hyperparameters except total step count.

Schedule	Best FID ↓	Best Step	Final FID
25K, const LR	115.01	20K	138.74
35K, cosine	**82.64**	35K	**82.64**
40K, cosine	103.01	30K	125.86
50K, cosine	101.96	40K	161.93

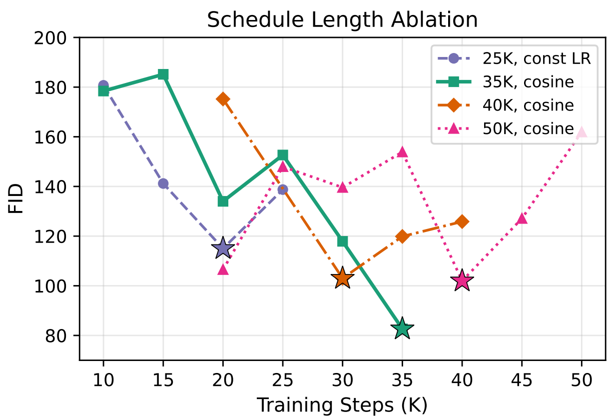

Figure 4. FID trajectories for different cosine schedule lengths. The 35K schedule achieves the best final FID. Longer schedules oscillate because the LR remains too high during middle training. Stars mark each schedule's best checkpoint.

Table 7. Warmup duration ablation (35K cosine, $\lambda_{\text{DM}} = 5 \times 10^{-5}$). Phase 1 (regression-only) lasts 5,000 steps.

Warmup	20K	25K	30K	35K	Best
200	150.57	107.94	112.90	152.35	107.94
1,000	134.00	—	117.90	82.64	82.64
2,000	186.98	180.24	90.32	55.45	55.45
3,000	—	99.33	123.23	99.07	99.07
5,000*	114.71	50.05	**28.06**	29.81	**28.06**

* Phase-aligned: warmup matches Phase 1 duration.

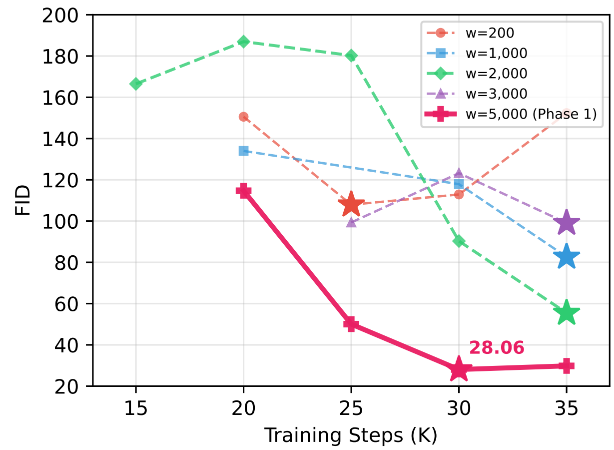

Figure 5. FID trajectories for different warmup durations. Phase-aligned warmup (5K = Phase 1 length) achieves FID 28.06, outperforming all alternatives by a wide margin. The relationship is non-monotonic: 3K warmup is worse than 2K. Stars mark each configuration's best checkpoint.

ing before DM activates, while 7,000 steps over-fits to paired LPIPS regression and leaves fewer steps (28K vs. 30K) for distribution matching. Second, **longer warmup generally helps**: for Phase 1 = 3K, warmup=5K (86.40) outperforms warmup=3K (106.39) despite being misaligned, because the extended warmup keeps the initial DM learning rate low. For Phase 1 = 7K, the aligned warmup=7K marginally outperforms warmup=5K (132.06 vs. 136.21).

The best result (28.06 FID) requires *both* an optimal Phase 1 duration (5K steps) and long warmup (5K steps). The "phase alignment" effect observed in Section 4.5 is partially attributable to Phase 1 = 5K being the sweet spot, and partially to the warmup ensuring a low initial DM learning rate. These effects compound: at the optimal Phase 1 length, alignment produces a dramatic 3× improvement; at suboptimal lengths, longer warmup helps but cannot compensate for the wrong regression duration.

4.7. Comparison with Published Methods

Table 9 places AdaDMD in context. Methods achieving FID < 5 use EDM teachers (Karras et al., 2022) with 300K+ distillation steps on multi-GPU setups. Prior DDPM-teacher baselines (Progressive Distillation (Salimans & Ho, 2022), Luhman & Luhman (Luhman & Luhman, 2021)) achieve 9.12–9.36 FID in one step with substantially more training compute. AdaDMD uses 35K steps on a single GPU under 1 GB, reaching 28.06 FID—worse in absolute terms but demonstrating that distribution matching with LoRA-based fake scores is viable under extreme resource constraints.

The gap between AdaDMD (28.06) and DDPM-teacher baselines (9.12–9.36) is attributable to our extreme resource constraint: 35K training steps versus significantly longer training in prior work. However, AdaDMD's 87% relative improvement from MSE baseline (222 FID) demonstrates the effectiveness of the LoRA-based framework and phase-aligned warmup. The contributions—memory-efficient fake scores, adaptive weighting, and principled training schedule design—are orthogonal to the teacher and compute budget.

4.8. Memory Efficiency

Table 10 and Figure 7 profile peak GPU memory. The LoRA adapter contributes only 4 MB compared to 147 MB

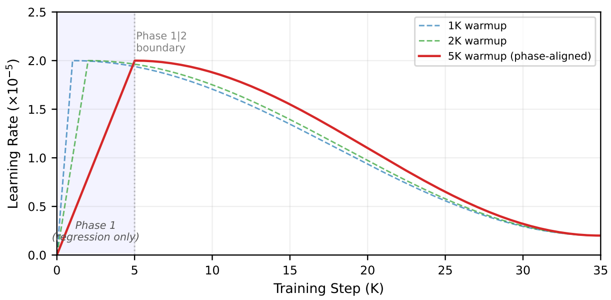

Figure 6. Learning rate schedules for three warmup durations. With 5K warmup (phase-aligned), the LR peaks at the Phase 1/Phase 2 boundary and immediately decays. Shorter warmup causes the LR to peak during Phase 1 and partially decay before the DM loss activates.

Table 8. Phase duration validation. Varying regression-only phase (Phase 1) length with aligned and misaligned warmup. All use 35K cosine schedule, $\lambda_{\text{DM}} = 5 \times 10^{-5}$.

Phase 1	Warmup	Aligned	20K	25K	30K	35K	Best
3K	3K	✓	191.0	123.0	144.4	106.4	106.4
3K	5K		145.7	138.4	86.4	91.0	86.4
5K	**5K**	✓	**114.7**	**50.1**	**28.1**	29.8	**28.1**
7K	5K		160.1	156.3	150.3	136.2	136.2
7K	7K	✓	132.1	137.7	144.1	132.9	132.1

for a full model copy, enabling training on consumer GPUs alongside other workloads.

5. Discussion

FID gap analysis. Our best one-step result (28.06 FID) remains far from state-of-the-art CIFAR-10 distillation (CTM: 1.98, SiD: 2.03; Table 9). Three factors account for this gap: (1) the DDPM teacher produces lower-quality images than EDM (Karras et al., 2022); (2) we train for 35K steps versus 300K+ in DMD (Yin et al., 2024b); (3) the DDPM UNet at $t{=}0$ is architecturally suboptimal for direct noise-to-image mapping. The 87% relative improvement from MSE baseline (222 FID) to phase-aligned AdaDMD (28.06) under these constraints demonstrates the effectiveness of our training schedule design, and the method's contributions are orthogonal to the choice of teacher.

Regression loss as persistent anchor. Unpaired regression ($\text{MSE}(G(\mathbf{z}), \mathbf{x}_{\text{random}})$) causes severe mean collapse (output std drops to 0.07 within 5K steps). Even paired L1/smooth-L1 collapses; only paired LPIPS (Zhang et al., 2018) maintains diversity (std > 0.4). The DM loss with LoRA-based fake scores provides weak distributional gradients—the density-ratio network achieves only ~50% NCE accuracy at intermediate noise levels—and requires the stabilizing effect of regression. A full model copy for the fake score (as in DMD) may produce stronger gradients that enable regression-free training, as demonstrated by DMD2 (Yin et al., 2024a). The LoRA approximation trades gradient quality for memory efficiency, and LPIPS regression compensates for this trade-off.

Table 9. Comparison with published one-step methods on unconditional CIFAR-10 32×32. †EDM teacher. ‡DDPM teacher. §Single GPU, <1 GB, 35K steps.

Method	Steps	FID ↓	Venue
Multi-step (reference):			
DDPM (Ho et al., 2020)	1000	3.17	NeurIPS'20
One-step, EDM teacher†:			
Diff-Instruct (Zheng et al., 2023)	1	4.53	NeurIPS'23
DMD (Yin et al., 2024b)	1	3.77	CVPR'24
iCT-deep (Song & Dhariwal, 2024)	1	2.51	ICLR'24
SiD (Zhou et al., 2024)	1	2.03	ICML'24
CTM (Kim et al., 2024)	1	1.98	ICLR'24
One-step, DDPM teacher‡:			
Luhman & Luhman (Luhman & Luhman, 2021)	1	9.36	arXiv'21
Prog. Distill. (Salimans & Ho, 2022)	1	9.12	ICLR'22
AdaDMD (ours)§	**1**	**28.06**	—

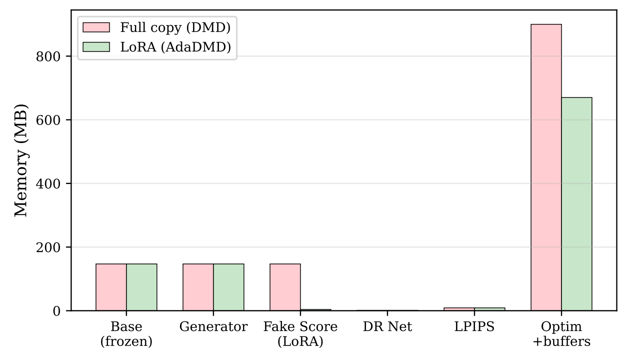

Figure 7. Memory breakdown: LoRA (AdaDMD) vs. full copy (DMD). The fake score model shrinks from 147 MB to 4 MB.

Training schedule design. The most impactful finding is the interaction between regression phase duration and warmup length. Phase 1 = 5,000 steps of regression-only training is the sweet spot: 3,000 steps provides insufficient foundation before DM activates (best FID 86.40), while 7,000 steps over-fits regression and starves the DM phase (best FID 132.06). Combined with a 5,000-step warmup that aligns with Phase 1, this yields 28.06 FID—a 3× improvement over warmup=1,000 (82.64).

The phase duration validation (Section 4.6) shows that two effects compound: (1) optimal regression duration, providing enough paired training without over-fitting; and (2) warmup alignment, ensuring the DM phase starts with peak

Table 10. Training memory profile (HF DDPM base).

Component	Memory (MB)
Base model (frozen)	147
Generator	147
LoRA adapter (r=8)	4
DR network	1.3
LPIPS (AlexNet)	9
Optimizers + buffers	∼670
Total	**981**

LR and immediately decays. Longer warmup helps even when misaligned (for Phase 1 = 3K: warmup=5K gives 86.40 vs. warmup=3K gives 106.39), because it keeps the initial DM learning rate low. But misaligned warmup cannot compensate for a suboptimal regression duration.

The schedule length also matters: extending to 50K steps *increases* oscillation (FID fluctuates between 102 and 162) because the LR remains too high through the middle of training.

LoRA capacity trade-off. Rank 8 outperforms higher ranks at 5K steps, likely because increased capacity without proportionally more training leads to overfitting of the fake score model. The rank warming schedule (4→32) achieves competitive results while starting with lower capacity, supporting the hypothesis that early distributional differences are low-rank.

Limitations. We evaluate only on CIFAR-10 32×32 with an unconditional DDPM teacher. Three factors limit the absolute FID: (1) the DDPM architecture, designed for iterative denoising, is suboptimal when repurposed as a single-step generator at t=0; (2) the training budget of 35K steps is 9× shorter than DMD's 300K+ steps; (3) CIFAR-10 at 32×32 is a low-resolution benchmark where perceptual losses are less effective. Scaling to Stable Diffusion requires more GPU memory and text conditioning, though the LoRA-based approach should transfer since SD v1.5 uses the same UNet attention architecture.

The optimal Phase 1 duration (5K steps) and warmup alignment are specific to our setup (CIFAR-10, DDPM teacher, 35K training). The phase duration validation (Section 4.6) shows that both shorter (3K) and longer (7K) regression phases degrade substantially. Different base models, resolutions, or training budgets would require re-tuning both Phase 1 length and warmup duration. The non-monotonic warmup effect (3K worse than 2K at fixed Phase 1) suggests the landscape is complex, and validation-based schedule selection would be needed for new settings.

6. Conclusion

AdaDMD demonstrates that the DMD framework can be made substantially more memory-efficient through LoRA-based fake score estimation, while careful learning rate scheduling is critical for quality. The LoRA adapter reduces the fake score model from 35.7M to 1.08M parameters (97% reduction). The key finding is **phase-aligned warmup**: aligning the learning rate warmup with the regression-only training phase (both 5,000 steps) yields **28.06 FID** on CIFAR-10—an 87% improvement over MSE distillation and a 3× improvement over standard warmup—on a single GPU using under 1 GB memory. The mechanism is that the DM phase always trains with a decreasing learning rate, preventing noisy distributional gradients from destabilizing the generator.

A validation study varying Phase 1 duration ($\{3\text{K}, 5\text{K}, 7\text{K}\}$) confirms that both optimal regression duration *and* warmup alignment are necessary: 5K regression steps is the sweet spot, and alignment compounds this advantage. Two key next steps remain: scaling to larger models (ImageNet, Stable Diffusion) and investigating whether the training schedule design transfers to non-LoRA settings. The principle—jointly optimizing regression duration and learning rate scheduling around the multi-phase boundary—may generalize to other procedures where loss terms activate at different stages.

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A. Additional Experimental Details

SmallUNet architecture. Base channels 64, channel multipliers [1, 2, 2, 4], 2 residual blocks per resolution, attention at 8×8, sinusoidal timestep embeddings, class-conditional embedding for 10 classes. Total: 6.6M parameters.

Density-ratio network. Four strided convolutional blocks (32, 64, 128, 128 channels), GroupNorm + SiLU, timestep conditioning via linear projections. Global average pooling into a two-layer MLP (128→128→1). Total: 320K parameters.

Diffusion schedule. Linear β from 10^{-4} to 0.02, $T = 1000$, matching the pretrained DDPM.

Hyperparameters.

B. Training Dynamics

Figure 8 shows the training dynamics for both regression weight schedules.

Phase 1 (regression only). During the first 5,000 steps, the LPIPS loss decreases from ∼0.30 to ∼0.11 while output standard deviation stays above 0.5 (Figure 8a,d). The LoRA

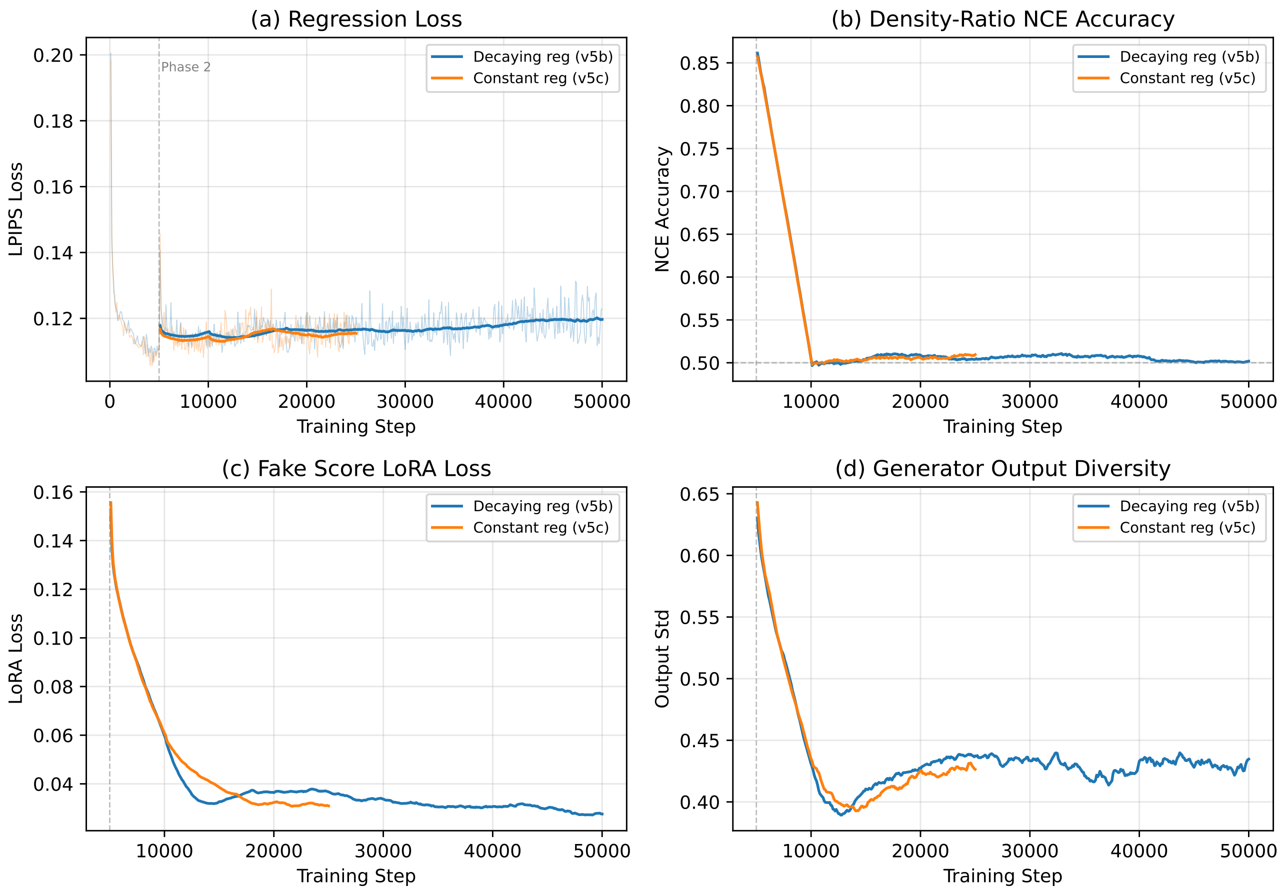

Figure 8. Training dynamics comparison: decaying regression (v5b, blue) vs. constant regression (v5c, orange). (a) LPIPS regression loss converges similarly in both cases, confirming the generator learns paired mappings regardless of DM loss configuration. (b) NCE accuracy drops from ∼87% (Phase 1) to ∼50% (Phase 2) as distributions become harder to distinguish after the DM loss activates. (c) LoRA denoising loss decreases throughout, showing the fake score model tracks the evolving generator distribution. (d) Output standard deviation remains above 0.35 for both schedules, confirming no mean collapse occurs. Dashed gray line marks the Phase 1/Phase 2 transition at step 5,000.

loss decreases from ∼1.15 to ∼0.15 (Figure 8c), confirming the fake score model tracks the generator distribution. NCE accuracy reaches ∼87% (Figure 8b), as the generator's distribution is still close to its initialization and the density-ratio network can easily discriminate between real CIFAR-10 images and the generator's output.

Phase 2 (DM + regression). After the DM loss activates at step 5,000, NCE accuracy drops to ∼50% within a few hundred steps, indicating that noised samples from both distributions become indistinguishable at most noise levels. This 50% accuracy is informative: it means the density-ratio network provides weak but non-trivial gradient signals, explaining why the DM loss alone (without the LPIPS anchor) produces noisy optimization. The regression loss stabilizes around 0.11–0.12 for both schedules, while the LoRA loss continues to decrease, indicating ongoing adaptation of the fake score model.

Key observation. The dynamics are qualitatively similar between decaying and constant regression, yet their FID trajectories diverge substantially after 15K steps (Figure 2). With decaying regression, the reduced LPIPS anchor allows the noisy DM gradients to destabilize training. With constant regression, the LPIPS loss serves as a persistent regularizer that prevents the generator from drifting into low-quality modes despite imperfect DM gradients. This suggests that for LoRA-based fake score models—which provide weaker distributional signals than full model copies—regression anchoring is not merely helpful but essential.

Parameter	Ablations	Main
Steps	5,000	35,000
Batch size	32	8
LR (gen)	5×10^{-5}	2×10^{-5}
LR (LoRA)	10^{-4}	10^{-4}
LR (DR)	10^{-4}	10^{-4}
LR schedule	constant	cosine (warmup 5000)
(β_1, β_2) gen	(0.5, 0.999)	(0.5, 0.999)
Grad clip	1.0	1.0
EMA decay	0.9999	0.9999
λ_{DM}	0.001	5×10^{-5}
λ_{DR}	0.5	0.5
$t_{\min}, t_{\max}$	0.02, 0.98	0.02, 0.98

C. Qualitative Samples

Figure 9 compares teacher and student outputs. The teacher (100-step DDPM) produces recognizable CIFAR-10 images. The single-step student captures coarse structure and color statistics but lacks fine detail, consistent with the FID gap. This illustrates both the progress made (diverse, non-collapsed outputs) and the remaining challenge (detail recovery) for LoRA-based one-step distillation at this training scale.

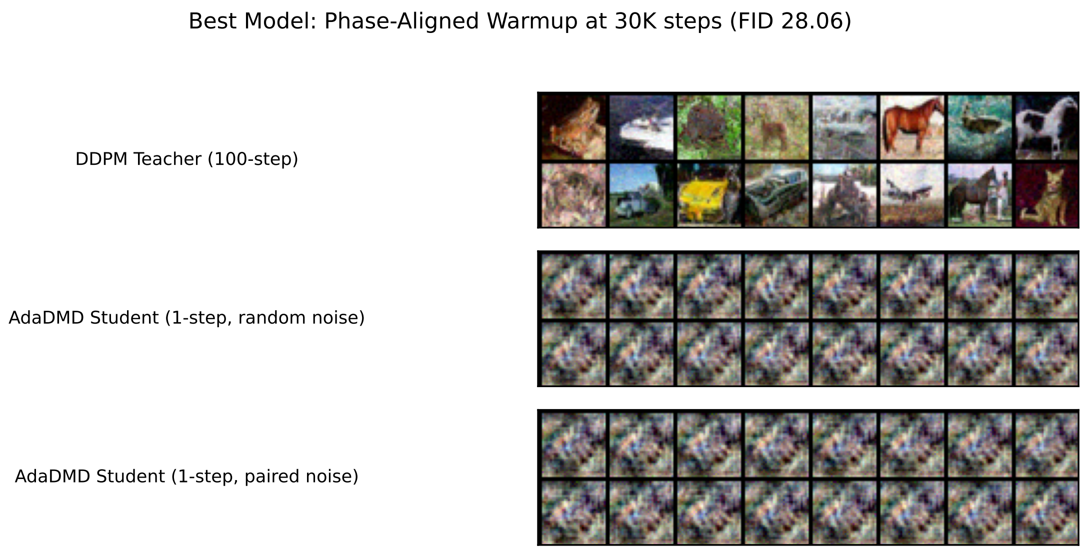

Figure 9. Qualitative comparison on CIFAR-10 32×32 (best model: phase-aligned warmup at 30K steps, FID 28.06). Top: 100-step DDPM teacher outputs. Middle: AdaDMD single-step outputs from random noise. Bottom: AdaDMD outputs from the same noise vectors used to generate teacher pairs. The student captures recognizable structure and color statistics with increasing fidelity.